

Lesson 2

Solution of the equations of motion for linear 1 d.o.f. and multi-d.o.f. systems

1 1 d.o.f. system

We consider the equation of motion for a linear 1 d.o.f. system, or the linearised equation obtained by linearising the motion for a general 1 d.o.f. system around a given equilibrium position. This takes the form of a linear, 2nd order differential equation with constant coefficients as follows:

$$m\ddot{x} + c\dot{x} + kx = f(t) \quad (1.1)$$

We assume in the following m , c and k to be positive valued. If we confine ourselves (cf. Lesson 1) to the case of velocity-dependent forces are of viscous nature and position-dependent forces of elastic nature, this is always true for m and c , whereas k is positive provided that the system is linear or is linearised around a stable equilibrium position.

We first consider the case of the homogeneous equation in the form (1.1), i.e. the case in which time-dependent forces are zero, known as the “free motion” of the system. Next, we consider the general case with non-zero $f(t)$, known as “forced motion”.

1.1 Free motion of the 1 d.o.f. system

We consider the 2nd order linear homogeneous differential equation with constant coefficients:

$$m\ddot{x} + c\dot{x} + kx = 0 \quad (1.2)$$

The general solution $x_g(t)$ for this equation is a combination of 2 linearly independent solutions having the form:

$$x = x_0 e^{\lambda t} \quad (1.3)$$

Deriving two times this solution and replacing in (1.2) we obtain:

$$(m\lambda^2 + c\lambda + k)x_0 = 0$$

and, in order to obtain non-trivial solutions ($x_0 \neq 0$) we prescribe:

$$m\lambda^2 + c\lambda + k = 0$$

and hence:

$$\lambda_{1,2} = -\frac{c}{2m} \pm \sqrt{\left(\frac{c}{2m}\right)^2 - \frac{k}{m}} \quad (1.4)$$

The solutions of (1.4) and hence the free motion of the system depend upon the amount of damping in the system.

We need to discuss three cases.



Case 1: $c=0$

In this case:

$$\lambda_{1,2} = \pm i\omega \quad \text{with} \quad \omega = \sqrt{\frac{k}{m}} \quad (1.5)$$

The general solution is hence a linear combination of two exponential functions having the general form (1.3) with exponents provided by (1.5):

$$x_g(t) = c_1 e^{-i\omega t} + c_2 e^{+i\omega t} \quad (1.6)$$

with c_1, c_2 arbitrary, complex-valued constants. To ensure that (1.6) is real valued, we prescribe these two constants to be complex conjugate:

$$\Rightarrow c_1 = \frac{A + iB}{2} \quad ; \quad c_2 = \frac{A - iB}{2}$$

We obtain:

$$\hookrightarrow x_g(t) = A \frac{e^{-i\omega t} + e^{+i\omega t}}{2} + B \frac{e^{i\omega t} - e^{-i\omega t}}{2i} = A \cos(\omega t) + B \sin(\omega t) \quad (1.7)$$

This solution can also be written as:

$$x_g(t) = C \cos(\omega t + \varphi) \quad \leftarrow x_g(\tau) \quad (1.8)$$

using the following relationships:

$$\begin{aligned} A &= C \cos(\varphi) & B &= -C \sin(\varphi) \\ C &= \sqrt{A^2 + B^2} & \varphi &= -\tan^{-1}\left(\frac{B}{A}\right) \end{aligned} \quad (1.9)$$

Solutions (1.7) and (1.8) show that the motion of a 1-d.o.f. undamped system is harmonic, with circular frequency ω ¹. This frequency does not depend upon the initial conditions applied on the system, and is therefore called the *natural frequency* of the undamped 1 d.o.f. system.

Coefficients A and B in equation (1.7) (or coefficients C and φ if solution (1.8) is used instead) are instead depending on the initial conditions applied on the system. Coefficients A and B can be derived by imposing the value of the initial position x_{t0} and velocity v_{t0} of the system at time $t=0$:

$$\begin{aligned} x_g(t=0) &= A = x_{t0} \\ \dot{x}_g(t=0) &= \omega B = v_{t0} \end{aligned}$$

In case the form (1.8) of the solution is used, coefficients C and φ are then derived using (1.9).

Case 2: $0 < c < 2m\omega$

We define *critical damping* the value:

$$c_c = 2m\omega$$

¹ We will hereafter denote by symbol ω the circular frequency expressed in *rad/s*. This is related to frequency f expressed in $\text{Hz}=\text{s}^{-1}$ by the relationship:

$$\omega = 2\pi f \quad f = \frac{\omega}{2\pi}$$

it is shown in this paragraph and the next that **this is the minimum amount of damping for which the motion of the 1 d.o.f. system becomes non-oscillatory**. We also define the **non-dimensional damping ratio** of the system h as:

$$h = \frac{c}{c_c} = \frac{c}{2m\omega}$$

In **this case**:

$$\lambda_{1,2} = -\alpha \pm i\omega_d \text{ with: } \alpha = -\frac{c}{2m} = -h\omega ; \omega_d = \sqrt{\frac{k}{m} - \left(\frac{c}{2m}\right)^2} = \sqrt{\omega^2 - (h\omega)^2} = \omega\sqrt{1-h^2} \quad (1.10)$$

Like in case 1, the **general solution is a linear combination of two exponential functions** with exponents provided by (1.10):

$$x_g(t) = e^{-\alpha t} (c_1 e^{-i\omega_d t} + c_2 e^{+i\omega_d t})$$

with c_1, c_2 arbitrary, complex-valued constants. Using the **same treatment introduced in case 1**, it is easy to show that this **solution can be transformed to the real-valued expression**:

$$x_g(t) = e^{-\alpha t} (A \cos(\omega_d t) + B \sin(\omega_d t)) = e^{-\alpha t} C \cos(\omega_d t + \varphi) \quad (1.11)$$

We see from (1.11) that the **free motion of a 'lightly damped'** (i.e. with damping below the critical value) **1 d.o.f. system is a decaying oscillation with circular frequency ω_d , different from the natural frequency of an undamped system** having the same mass and stiffness properties. We note however that is the amount of damping in the system is much lower than critical (i.e. $h=0.01$), based on (1.10) the **difference between the natural frequency of the damped and undamped system is negligible**.

Case 3: $c > 2m\omega$

In this case, the **damping in the system is above the critical value**, $h > 1$. Based on (1.4), we note that **in this case the two exponents $\lambda_{1,2}$ will be real-valued and negative**. Indeed, the argument of the square root is positive (hence the root is real) and the root value is less than $c/2m$, because k and m are constant parameters. **We have** therefore:

$$\lambda_1 = -\alpha_1 ; \quad \lambda_2 = -\alpha_2 \quad (1.12)$$

with α_1, α_2 constant, real positive values.

Hence, **the solution for this case is a linear combination of two exponential functions** with time – varying real negative exponent.

$$x_g(t) = Ae^{-\alpha_1 t} + Be^{-\alpha_2 t} \quad (1.13)$$

This produces a **non-oscillatory decaying motion**.

1.2 **Forced motion** of the 1 d.o.f. system

We consider now the **complete, non-homogeneous 2nd order linear differential equation**:

$$m\ddot{x} + c\dot{x} + kx = f(t)$$

The **solution for this equation is the sum of the general solution $x_g(t)$ of the homogeneous equation (1.2) summed with one 'particular' solution of the complete equation, denoted as $x_p(t)$** :

$$x(t) = x_g(t) + x_p(t) \quad (1.14)$$

The term $x_g(t)$ was determined in the previous sub-section, so is known. We note furthermore that in presence of damping this term will be decaying in time (either with or without oscillations), and will become negligible after a sufficiently long time. Hence we call $x_p(t)$ the “stationary” vibration of the system, and we will be normally most interested in this part of the solution. In order to evaluate $x_p(t)$ we need to specify the type of time-dependent force acting on the system. We consider below the particular but very important case of a harmonic force:

$$f(t) = F \cos(\Omega t) \quad (1.15)$$

with Ω the circular frequency of the time-dependent force. In order to find the particular solution for the case considered here, it is convenient to transform equation (1.14) into a complex-valued form. To this end, we consider two equations, one with harmonic forcing of cosine type and the other with sine-type forcing function. We denote by x and y the unknowns in the two equations:

$$\begin{cases} m\ddot{x} + c\dot{x} + kx = F \cos(\Omega t) \\ m\ddot{y} + c\dot{y} + ky = F \sin(\Omega t) \end{cases}$$

next, we multiply the second equation by the imaginary unit i and sum the two equations. Defining a complex-valued unknown x^* :

$$x^* = x + iy$$

and recalling:

$$\cos(\Omega t) + i \sin(\Omega t) = e^{i\Omega t}$$

we get:

$$m\ddot{x}^* + c\dot{x}^* + kx^* = Fe^{i\Omega t} \quad (1.16)$$

A particular solution for this complex-valued equation is in the form:

$$x^* = Xe^{i\Omega t} \quad (1.17)$$

with x_0 complex-valued. Deriving with respect to time (1.17) and replacing in (1.16) we get:

$$(-\Omega^2 m + i\Omega c + k)X = F \quad (1.18)$$

and hence:

$$X = \frac{1}{-\Omega^2 m + i\Omega c + k} F = G(i\Omega) F \quad (1.19)$$

with $G(i\Omega)$ the frequency response function (FRF) of the 1 d.o.f. linear system. Since x_0 is a complex value, it can be represented using the ‘polar’ form as:

$$X = |X| e^{i\psi}$$

with ψ the ‘phase’ of the complex number. Using this result, we obtain the particular integral $x_p(t)$ of the real-valued equation (1.15) as:

$$x_p(t) = \operatorname{Re}(Xe^{i\Omega t}) = \operatorname{Re}(|X|e^{i\psi}e^{i\Omega t}) = \operatorname{Re}(|X|e^{i(\Omega t + \psi)}) = |X| \cos(\Omega t + \psi)$$

We see that the ‘stationary response’ of the linear 1 d.o.f. system to a harmonic force is a harmonic motion having the same circular frequency as the force, but presenting a phase shift with respect to the force. Zero phase shift means the maximum positive displacement occurs at the same time as the maximum positive force (we say in this case that force and motion are in phase), a phase shift of

$\pm\pi$ means that the maximum positive displacement occurs when the force value is at a minimum (we say in this case that force and displacement are in **counter-phase** or in phase opposition). The amplitude of the system's stationary response under a given harmonic force depends from the product of the amplitude of the force times the magnitude of the FRF, whereas the phase shift ψ between the force and the system's response is given by phase angle of the complex-valued FRF. Figure 1 below illustrates the **Bode diagram** (using linear scales on both axes) for the 1-d.o.f. system. The dashed line is for zero damping ($c=0$), whereas different solid lines are plotted for different cases of non-zero damping ($h = \frac{c}{c_c} = 0.05, 0.1, 0.707$).

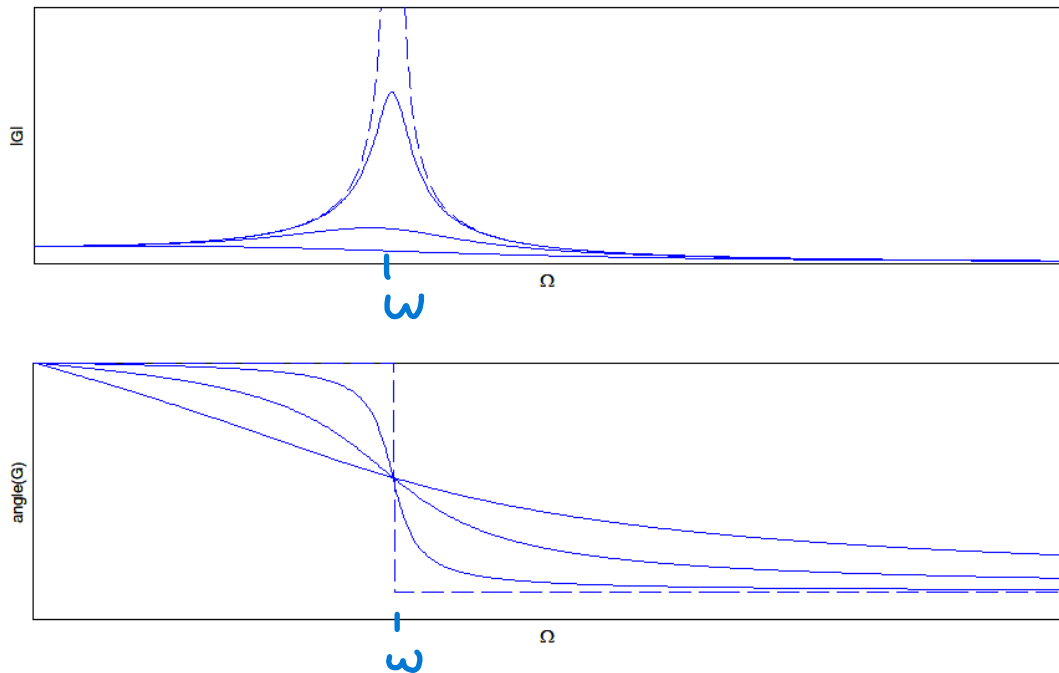


Figure 1: Bode diagram for the 1-d.o.f. system for different values of $h = \frac{c}{c_c}$

The plot in Figure 1 shows that **three distinct frequency ranges are relevant to the trend of the system's FRF**: the **quasi-static range** (frequency of excitation Ω is well below the system's undamped natural frequency ω), the **seismographic** or **suspended range** (frequency of excitation Ω is well above the system's undamped natural frequency ω) and the **resonance** condition (frequency of excitation Ω is close to the system's undamped natural frequency ω). In order to obtain a physical interpretation of the system's behaviour in these three frequency ranges we rewrite Eq. (1.18) in the form:

$$Fe^{i\Omega t} + \Omega^2 m X e^{i\Omega t} - i\Omega c X e^{i\Omega t} - k X e^{i\Omega t} = 0 \quad (1.20)$$

Each term in Eq. (1.20) is a **phasor**, i.e. a vector in the complex plane having constant magnitude and rotating with angular speed Ω which is the frequency of the force exciting the vibration of the system. Eq. (1.20) can be interpreted as a balance between the phasor representing the time dependent force (the first one in the equation) and the other three phasors representing respectively inertial effects, viscous damping effects and elastic effects in the system.

In the **quasi-static range**:

$$\Omega^2 \ll \omega^2 = \frac{k}{m} \rightarrow m\Omega^2 \ll k \quad (1.21)$$

Assuming **small damping**, the **phasors representing inertial and damping effects are both negligible compared to the one representing elastic effects**, hence the **time-dependent excitation applied on the system is almost exclusively balanced by the elastic effect**. As a consequence, the FRF takes values which are close to real positive numbers (phase angle is close to 0) which are almost not affected by the frequency of excitation (i.e. in this range both the magnitude and phase of the FRF are almost constant for varying excitation frequency). It is often said that **in the quasi-static frequency range the motion of the system can be represented as a “sequence of static/equilibrium states” as if the force was varying so slowly that any dynamic effect can be neglected**.

In the **seismographic / suspended range**, reversing the inequality in eq. (1.21) and again assuming **small damping** it can be concluded that the **time dependent excitation is mostly balanced by the inertial effect**. Because this is the **product of mass and acceleration** and because the system's **acceleration for harmonic motion is proportional to the square of the circular frequency**, the amplitude of the FRF in this range is inversely proportional to the square of the frequency of excitation Ω . Furthermore, in order to meet the force balance expressed by Eq. (1.20) **the inertial force needs to be opposite to the external excitation**. But the inertial force is opposite to the system's acceleration which in turn, for harmonic motion, is in counterphase (180 deg out of phase) to the displacement of the system, which ultimately means, the relative phase of the system's displacement relative to the time dependent excitation (which is the phase angle of the FRF) is in this range -180 Deg or $-\pi$.

In the **resonance condition**:

$$\Omega^2 = \omega^2 = \frac{k}{m} \rightarrow m\Omega^2 = k \quad (1.22)$$

hence the **two phasors representing inertial and elastic effects cancel with each other and the only effect providing balance to the time dependent excitation is the damping one**. However, if the amount of damping is low (it can be shown this actually depends on the damping ratio $h = \frac{c}{c_c}$) a

large displacement x_0 is needed in Eq. (1.20) in order to balance the time dependent force. Therefore, **for low values of damping ratio a large peak** (known as the *resonance peak*) **is observed in the FRF**. In the limit case in which $c=h=0$, the resonance peak actually becomes a vertical asymptote, and the FRF is not defined in the resonance condition.

2 **Multi-d.o.f. systems**, Solution for the **free and undamped motion**.

We now switch to the case of a **linear** (or **linearised**) **multi-d.o.f. system**. In this case the **equations of motion take the form matrix form**:

$$[m]\ddot{x} + [c]\dot{x} + [k]x = f(t) \quad (2.1)$$

The first case we take in consideration is that of the **free** (i.e. **not forced, $f(t)=0$**) and **undamped** (i.e. **$[c]=[0]$**) case:

$$\Rightarrow [m]\ddot{x} + [k]x = 0 \quad (2.2)$$

This case is particularly important for at least three reasons:

- 1) the solution of the free and undamped case is the **most easy to find**, and thus plays an introductory role with respect to more complicated cases;
- 2) as will be shown by the lesson on the so called **modal superposition approach**, a **system with low level of damping behaves in a way that is very similar to that of an undamped system**, so the **undamped case is also useful for understanding the actual behaviour of systems with low damping**;

- 3) the **results** obtained by solving equation (2.2), **namely the** so called **natural frequencies** and the related **modes of vibration of the system** are **needed to define the “modal superposition approach”** (see lesson 3) that is **one of the most useful conceptual tools to understand the actual behaviour of a complicate, multi-dof vibrating system even under the presence of damping and forcing effects.**

In order to solve equation (2.2), we observe that **this equation is the matrix form for a system of n second order differential equations**, and that this **system is linear with time-independent coefficients**. Thus, the **solution** of this equation is a **linear combination of $2n$ solutions** in the exponential form:

$$\underline{x} = \underline{x}_0 e^{\lambda t} \quad (2.3)$$

where in general $\underline{x}_0$ and λ are **complex-valued constants**, so that in particular the constant λ in the exponent is composed by a real part α and an imaginary part ω :

$$\lambda = \alpha + i\omega$$

Furthermore, it is **possible to demonstrate that if the mass and stiffness matrices are:**

- i) **symmetric and**
- ii) **positive definite**

then the **real part in the constant lambda drops down to zero**, so that the constant at exponent in the **solution** (2.3) becomes:

$$\underline{x} = \underline{x}_0 e^{i\alpha t} \quad (2.4)$$

In the case we are treating, we observe that **the two conditions i) and ii) above are certainly satisfied by the mass and stiffness matrices** of the system. Indeed, matrices $[m]$ and $[K]$ are **symmetric by definition**. If we consider for instance the mass matrix we have:

$$[m] = [\lambda_m(0)]^T [m_{ph}] [\lambda_m(0)]$$

with $[m_{ph}]$ being a diagonal (and thus symmetric) matrix. Therefore:

$$[m]^T = [\lambda_m(0)]^T [m_{ph}]^T [\lambda_m(0)] = [\lambda_m(0)]^T [m_{ph}] [\lambda_m(0)] = [m]$$

The same two matrices are **also definite positive**, in fact we recall that a matrix $[A]$ is said to be definite positive if:

$$\underline{x}^T [A] \underline{x} > 0 \quad \text{for any } \underline{x} \neq \underline{0}$$

now we **observe that:**

- i) since the **kinetic energy of the system must be strictly greater than zero** for any movement of the system except $\dot{\underline{x}} = \underline{0}$ (which means the still system), **then** $T = \frac{1}{2} \dot{\underline{x}}^T [m] \dot{\underline{x}} > 0$ for any $\dot{\underline{x}} \neq \underline{0}$, which **means $[M]$ is definite positive.**
- ii) since a **theorem exists** that **states that “given a system subjected to conservative forces, an equilibrium position will be stable if and only if the total potential energy has a minimum value in that position”** then $V = \frac{1}{2} \underline{x}^T [k] \underline{x}$ **must be strictly positive** for any value of the position $\underline{x}$, except $\underline{x} = \underline{0}$, which represents the equilibrium position.

so, **in conclusion**, we may state that for our system the **solution of equation** (2.3) will be in the form of the linear combination of $2n$ solutions of the type (2.4). To find these solutions, we simply take the first and second order time derivatives of solution (2.4):

$$\underline{x} = \underline{x}_0 e^{i\omega t} \quad \dot{\underline{x}} = i\omega \underline{x}_0 e^{i\omega t} \quad \ddot{\underline{x}} = (i\omega)^2 \underline{x}_0 e^{i\omega t} = -\omega^2 \underline{x}_0 e^{i\omega t}$$

and we introduce these expressions into equation (2.1), that by obvious re-arranging becomes:

$$(-\omega^2 [m] + [k]) \underline{x}_0 = \underline{0} \quad (2.5)$$

where the coefficient $e^{i\omega t}$ has been simplified since it is always different from zero (it is indeed a complex number with unit modulus).

The above equation defines the solutions to equation (2.2) and takes the form of a system of linear equations in matrix form. This system is “homogeneous”, which means that the constant terms at the right hand side are all zero-valued. If matrix $(-\omega^2 [m] + [k])$ is non-singular, one and only one solution exists for (2.5), consisting of the so-called *trivial solution* $\underline{x}_0 = \underline{0}$ which represents the system keeping steady at the equilibrium position. We may however look for *non-trivial solutions*, i.e. actually representing a motion of the system, by prescribing:

$$\det(-\omega^2 [m] + [k]) = 0 \quad (2.5b)$$

Since the mass and stiffness matrices are square matrices with order n , it is easy to find that equation (2.5) takes the form of a polynomial expression in ω , with order $2n$ which will provide $2n$ solutions in the form:

$$\left\{ \begin{array}{l} \omega = \pm \omega_1 \\ \omega = \pm \omega_2 \\ \dots \\ \omega = \pm \omega_n \end{array} \right.$$

where the real values $\omega_1, \omega_2, \dots, \omega_n$ take the name of “natural angular frequencies” of the system.

Now, coming back to the solution of equation (2.4), if we replace in that equation the generic value of ω with ω_j , being ω_j one of the values of the natural angular frequencies $\omega_1, \omega_2, \dots, \omega_n$, since the matrix of coefficients $(-\omega^2 [m] + [k])$ becomes singular, we must conclude that one of the equations in the system becomes a linear combination of the other $n-1$ equations, and we thus may delete this equation that does not add any additional information on the solution with respect to the other. By this procedure, we remain with an underdetermined system of equations, with $n-1$ equations in n unknowns. To solve this system, we arbitrarily set to 1 the value of one of the unknowns in vector $\underline{x}_0$ and by this way we determine a univoque solution for equation (2.5). This solution, that will be denoted by $\underline{x}_0^{(j)}$ has the following two properties:

- it is defined with *arbitrary amplitude*: indeed, by considering the way we have found the solution it is easy to see that if $\underline{x}_0^{(j)}$ is a solution of equation (2.5), then for any constant value c different from zero also $c \underline{x}_0^{(j)}$ is solution of the same equation.
- it is associated to the value ω_j of the natural angular frequency that was substituted into equation (2.4): indeed it is easy to see that by using a different value of ω_j a different solution would be obtained, this is the reason why we are using superscript (j) to denote this vector.

We are now finally able to define the motion of the free undamped system by combining the $2n$ solutions we have found by the above described procedure: thus the general solution $\underline{x}_g(t)$ for equation (2.3) is:

$$\underline{x}_g(t) = \sum_{j=1}^n \underline{x}_0^{(j)} (a_{1j} e^{-i\omega_j t} + a_{2j} e^{i\omega_j t}) \quad (2.6)$$

where $a_{11}, a_{21}, \dots, c_{1n}, c_{2n}$ are $2n$ complex-valued coefficients of the linear combination. Since we are seeking for a real-valued solution $\underline{x}_g(t)$, we may introduce the same treatment as for the 1 d.of. system, cf. Section 1.1, and reduce (2.6) to the real-valued expression:

$$\underline{x}_g(t) = \sum_{j=1}^n \left(\underline{x}_0^{(j)} (A_j \cos(\omega_j t) + B_j \sin(\omega_j t)) \right) = \sum_{j=1}^n \left(\underline{x}_0^{(j)} C_j \cos(\omega_j t + \varphi_j) \right) \quad (2.7)$$

We see from equation (2.7) that the free undamped motion of the system is a combination of n harmonic components of motion, where n is the number of degrees of freedom of the system. Each of these harmonic components is characterised by a specific angular frequency ω_j and by a “shape” of the motion (which is described by the mode of vibration $\underline{x}_0^{(j)}$, as explained in detail in subsection 2.1). Frequency and “shape” of the motion components are intimately related, so that each shape of the free motion is possible only if that motion happens at the corresponding natural frequency.

Furthermore, looking at equation (2.7) we observe that:

- $\omega_1, \omega_2, \dots, \omega_n$ are known quantities, indeed, they are the “natural frequencies” of the system and, recalling that they are obtained by solving equation (2.5), it is clear that their value is determined by the inertial and stiffness properties of the system, which finally means the values inside the mass and stiffness matrices;
- in the same way, $\underline{x}_0^{(1)}, \underline{x}_0^{(2)}, \dots, \underline{x}_0^{(n)}$ are known quantities (the “modes of vibration of the system”), and also depend upon the mass and stiffness properties of the system;
- the constants $A_1, B_1, \dots, A_n, B_n$ (or alternatively $C_1, \varphi_1, \dots, C_n, \varphi_n$) defining the solution do not depend on the physical parameters of the system, but instead on the initial conditions assigned to the system. In other words, different combinations of these coefficients describe different possible motions of the system that are produced by different combinations of initial positions and speed of the system coordinates (see section 2.4)

2.1 Physical interpretation of the modes of vibration

In order to complement about the physical interpretation of solution (2.7) for the free undamped motion of a n -dof linear system, we concentrate in this subsection on the physical meaning of the “modes of vibration”. To this end, we take as an easy example the 2-dof system represented in figure 2.1 below:

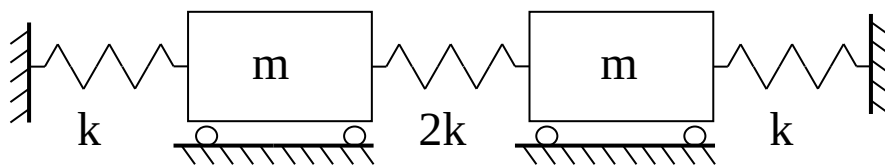


Figure 2.1: 2-dof system formed by two masses moving in the horizontal direction

In order to describe the configuration of the system we use the following two independent coordinates:

- x_1 horizontal displacement of the left body and
- x_2 horizontal displacement of the right body

We will study in detail this system in the next lesson, where it will be demonstrated that the two natural frequencies are:

$$\omega_1 = \sqrt{\frac{k}{m}} \quad ; \quad \omega_2 = \sqrt{\frac{5k}{m}}$$

and the related modes of vibration are:

$$\underline{x}_0^{(1)} = \begin{Bmatrix} 1 \\ 1 \end{Bmatrix} \quad ; \quad \underline{x}_0^{(2)} = \begin{Bmatrix} 1 \\ -1 \end{Bmatrix}$$

if we use these results and the general matrix expression of the free undamped motion (eq. 2.6) to re-write in scalar form the motion of the 2 mass system, we get:

$$x_1(t) = C_1 \cos\left(\sqrt{\frac{k}{m}}t + \varphi_1\right) + C_2 \cos\left(\sqrt{\frac{5k}{m}}t + \varphi_2\right)$$

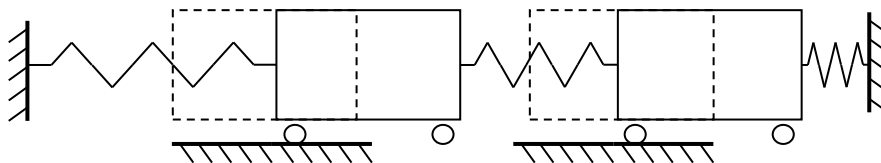
$$x_2(t) = C_1 \cos\left(\sqrt{\frac{k}{m}}t + \varphi_1\right) - C_2 \cos\left(\sqrt{\frac{5k}{m}}t + \varphi_2\right)$$

as anticipated, the time history of the two independent coordinates is the sum of two harmonic components. If we confine our attention to the first harmonic component (e.g. we set for the moment $C_2=0$) we get:

$$x_1(t) = C_1 \cos\left(\sqrt{\frac{k}{m}}t + \varphi_1\right)$$

$$x_2(t) = C_1 \cos\left(\sqrt{\frac{k}{m}}t + \varphi_1\right) = x_1(t)$$

so, we see that in the first harmonic component, the one with the lowest natural frequency, the two masses are moving exactly in the same way, that means they are moving in phase with the same amplitude. If during this type of motion of the system we take a “snapshot at a given time, we get the following picture (the dashed boxes represent the two masses in their equilibrium position):



thus, we see that in this component of the motion the masses are oscillating together, without a relative displacement between the two bodies, so that the central spring is neither extended nor compressed.

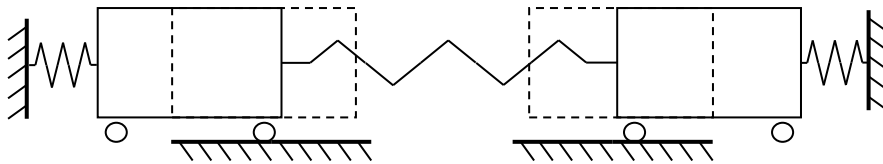
This “shape” of the motion is completely described by vector $\underline{x}_0^{(1)}$ representing the first mode of vibration: indeed, the equal amplitude of the motion for the two coordinates is represented by the equal absolute value of the two elements of vector $\underline{x}_0^{(1)}$, while the fact that the two masses are moving in phase one to the other is indicated by the fact that the two elements in vector $\underline{x}_0^{(1)}$ have the same sign.

We now come to the **second harmonic component of motion for our example**. If we now set $C_1=0$ and $C_2 \neq 0$ we get:

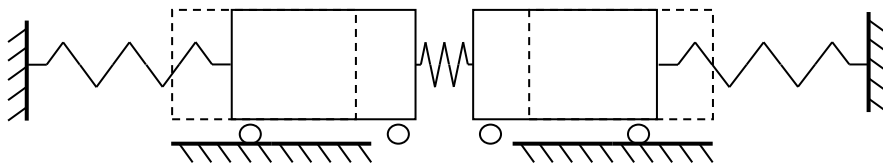
$$x_1(t) = C_2 \cos\left(\sqrt{\frac{5k}{m}}t + \phi_2\right)$$

$$x_2(t) = -C_2 \cos\left(\sqrt{\frac{5k}{m}}t + \phi_2\right) = -x_1(t)$$

in this second motion component, the **motion of the body at the right side is exactly opposite to that of the left body**: this **corresponds to a totally different shape of motion**, where the **two bodies start their oscillation by moving away one from the other and then**, after $\frac{1}{2}$ cycle **invert their motion** and **start coming near one to the other**. In a more precise way, we may state that **the motion of the two bodies have the same amplitude**, but **opposite phase**. If we take again a snapshot of this particular kind of motion, during the first part of the oscillation (i.e. when the masses are moving away one from the other) we get:



while a snapshot taken during the phase of the motion when the two masses are getting near one to the other will result in the following:



Again, we observe that the **shape of the motion is fully described by the corresponding mode of vibration** $\underline{x}_0^{(2)}$ where the **information about the equal amplitude for the motion of the two masses is given by the same absolute value of the two elements in the vector**, while the **information about the opposite phase of the motion is given by the opposite sign of the two elements in the vector**.

Based on the above example, we may now generalise the **physical interpretation of the “modes of vibration” vectors**, by stating that, **given any mode of vibration vector** and considering any couple of elements in the vector:

- the **ratio of the absolute values of the two elements represents the ratio of the amplitudes of motion for the two corresponding coordinates** in the system.
- the **relative sign represents the relative phase of motion between the two corresponding coordinates**, a **positive relative sign denoting a motion in phase** and a **relative negative sign denoting a motion in counterphase**.

We also note by the way that **in the free undamped motion of a multi-dof system**, since the **modes of vibrations are represented by real-valued vectors**, the **different coordinates may only vibrate exactly in phase or exactly in counterphase one to the other**.

2.2 The **modal matrix**

We have seen above that a **system with n dofs will have exactly n modes of vibration**, each one defined by a column vector $\underline{x}_0^{(j)}$ **formed by n elements**. The **whole information about the modes** of

vibration of the system may thus be collected in the so-called “modal matrix” $[\Phi]$ where the j -th column corresponds to the j -th mode of vibration $\underline{x}_0^{(j)}$:

$$[\Phi] = \begin{bmatrix} \underline{x}_0^{(1)} & \underline{x}_0^{(2)} & \dots & \underline{x}_0^{(n)} \end{bmatrix} \quad (2.8)$$

it is clear from this definition that the modal matrix $[\Phi]$ is a square $n \times n$ matrix. As we will see later in this lesson and in the next lesson, this matrix has a big importance in the study of the motion of the system, not only in the case of the free undamped motion, but also in the case of the damped and/or forced motion (by using the modal superposition principle).

2.3 A more efficient way to find the natural frequencies and modes of vibration of the system

In section 2 a mathematical procedure to compute the natural frequencies and related modes of vibration of a multi-dof system has been described. This procedure is the most straightforward from the conceptual point of view, but it is not the most efficient from the point of view of ease of use.

Indeed, given the mass and stiffness matrices of the system, one should write down equation (2.5) and solve it, which is an acceptable task (may be solved by hand) for a two-dof system, but rapidly becomes impossible to manage as soon as the number of dofs of the system increases.

If we come back to equation (2.4), it is easy to see that the problem of defining the natural frequencies and modes of vibration may be solved in a much easier way by re-stating the problem as an eigenvalue-eigenvector search. To this end, we recall first that, given a square matrix $[A]$, the eigenvalues λ and eigenvectors $\underline{X}$ are the solutions of the following problem:

$$[A]\underline{X} = \lambda \underline{X} \quad \text{or} \quad (\lambda[I] - [A])\underline{X} = 0 \quad (2.9)$$

where $[I]$ is an identity matrix with the same dimension as $[A]$.

Then, we rewrite equation (2.4) by pre-multiplying both sides by the inverse of the mass matrix². The modified version of equation (2.4) that we get is:

$$\left((-\omega^2 [I] + [m]^{-1} [k]) \underline{x}_0 = 0 \right) \quad (2.4b)$$

By comparing eq. (2.4b) with eq. (2.9) we see that:

- the eigenvalues of the matrix $[m]^{-1} [k]$ are the square of the system natural circular frequencies ω_j , for this reason the natural frequencies of the system are also called the “eigenfrequencies”;
- the eigenvectors of the matrix $[m]^{-1} [k]$ are the system modes of vibration $\underline{x}_0^{(j)}$, for this reason often the modes of vibration are directly called the “eigenvectors” or “eigenmodes” of the system;

This simple observation gives us a very powerful tool to find the natural frequencies and modes of vibration, even for systems with many dofs. Indeed, well established numerical methods exist to find the eigenvalues and eigenvectors of square matrices, and these methods are available in several numerical libraries and packages, like e.g. MATLAB®. Thus, this property will be used throughout the course, especially during numerical exercises and year work.

By the way, having recognised that the modes of vibration of the system are the eigenvectors of matrix $[m]^{-1} [k]$ we observe that since a theorem states that “the eigenvectors of a matrix are linearly independent”, then the modal matrix defined by equation (2.8) is for sure a non-singular

² This is indeed possible since as explained before the mass matrix is positive definite and thus for sure it is non singular

matrix. This property is indeed **important** and will be used in the next subsection as well as in Lesson 3.

2.4 **Response to initial conditions**

Until now, we have concentrated our attention on the **general features** (shape, frequency) of the different motion components, but we have not yet dealt with the **problem of finding the actual motion of the system corresponding to a given set of initial conditions**.

By **initial conditions**, we mean that **for each independent coordinate** of the system, an **initial position and an initial velocity** (value of the time derivative) are **prescribed at time $t=0$** . From a mathematical point of view, this means that

$$\left\| \begin{array}{l} \underline{x}_g(t=0) = \underline{x}_{t0} \\ \dot{\underline{x}}_g(t=0) = \underline{v}_{t0} \end{array} \right\| \quad (2.10)$$

where $\underline{x}_{t0}$ and $\underline{v}_{t0}$ are the **vectors collecting the prescribed values for the initial positions and velocities** of all system degrees of freedom.

We now **take the general solution of the free undamped motion** in the real-valued form (2.7) (using the first of the two forms of the solution, i.e. the one featuring the sum of one cosine and one sine functions) and we introduce in that expression conditions (2.10).

The time derivative of the solution (2.7) is:

$$\dot{\underline{x}}_g(t) = \sum_{j=1}^n \left(\underline{x}_0^{(j)} \omega_j \left(-A_j \sin(\omega_j t) + B_j \cos(\omega_j t) \right) \right) \quad (2.11)$$

Replacing (2.7) and (2.11) in (2.10) **we get:**

$$\left\| \begin{array}{l} \sum_{j=1}^n \left(\underline{x}_0^{(j)} A_j \right) = \underline{x}_{t0} \\ \sum_{j=1}^n \left(\underline{x}_0^{(j)} \omega_j B_j \right) = \underline{v}_{t0} \end{array} \right\|$$

By **defining the two vectors $\underline{A}$ and $\underline{B}^*$** :

$$\underline{A} = \{A_1 \quad A_2 \quad \dots \quad A_n\}^T$$

$$\underline{B}^* = \{\omega_1 B_1 \quad \omega_2 B_2 \quad \dots \quad \omega_n B_n\}^T$$

and recalling from (2.8) the definition of the modal matrix, the **two matrix equations above may be written as:**

$$\left\{ \begin{array}{l} [\Phi] \underline{A} = \underline{x}_{t0} \\ [\Phi] \underline{B}^* = \underline{v}_{t0} \end{array} \right. \quad (2.12)$$

The first equation in (2.12) provides the $A_1, \dots, A_n$ coefficients as function of the given initial positions, the second one provides vector $\underline{B}^*$ (and then by obvious calculations coefficients $B_1, \dots, B_n$) for a **given set of initial velocities of the system**. Since as observed in the previous subsection the modal matrix $[\Phi]$ **is non-singular**, the **solution for the two above linear problem exists and is unique**.

Once parameters $A_1, \dots, A_n, B_1, \dots, B_n$ are known, the motion of the system for any independent coordinate $x_1, x_2, \dots, x_n$ and for any value of time t may be directly computed using equation (2.7).

3 Multi-d.o.f. systems. Solution for the free damped motion

A second case that has to be considered in the study of the solution of the equations of motion for a linear(ised) multi-dof system is the case of the free damped motion, which means that external forcing effects are set to zero as in the previous section, but the effect of damping is no more neglected.

The corresponding equation in matrix form is thus:

$$[m]\ddot{\underline{x}} + [c]\dot{\underline{x}} + [k]\underline{x} = \underline{0} \quad (3.1)$$

where we assume that also the damping matrix $[c]$, as the mass and stiffness matrices, is positive definite. This mathematical property corresponds to the fact that the viscous dampers placed in the system are able to dissipate energy under any possible motion of the system. A mechanical system having such a property is called a *dissipative system*.

We define again as the general solution for equation (3.1) a linear combination of $2n$ functions of the type:

$$\underline{x} = \underline{x}_0 e^{\lambda t} \quad (3.2)$$

but this time, the exponent λ in the solution will have a non-zero real part:

$$\lambda = \alpha + i\omega$$

to determine the $2n$ exponents λ and thus the solution of equation (3.1) we take the first and second derivative wrt time of the solution (3.2):

$$\dot{\underline{x}} = \lambda \underline{x}_0 e^{\lambda t} \quad : \quad \ddot{\underline{x}} = \lambda^2 \underline{x}_0 e^{\lambda t} \quad (3.2)$$

and we substitute into eq. (3.1) that, after obvious re-arranging of the terms takes the form:

$$(\lambda^2 [m] + \lambda [c] + [k]) \underline{x}_0 = \underline{0} \quad (3.3)$$

as for equation (2.5) in the case of the undamped system, we must seek for non-trivial solutions for the homogeneous problem (3.3), and this reduces to the condition:

$$\det(\lambda^2 [m] + \lambda [c] + [k]) = 0 \quad (3.4)$$

which provides $2n$ (generally complex-valued) exponents λ_j and for each of them a corresponding non-trivial solution $\underline{x}_0^{(j)}$.

Without adding too much complication with mathematical details, we recall here that, just as for the undamped free motion of the system, it is possible to re-formulate equation (3.3) in the form of an eigenvalue/eigenvector problem but, differently from the case of the undamped system, this requires to re-write the equations of motion (3.1) in the form of (homogeneous) state equations:

$$\dot{\underline{z}} = [A]\underline{z}$$

where $\underline{z}$ is the state vector, composed by the n coordinates $\underline{x}$ and the n first order time derivatives of the coordinates (velocities) $\dot{\underline{x}}$, while $[A]$ is a $2n \times 2n$ state matrix. Details on this procedure and on the calculation of the complex exponents, related eigenvectors and derivation of the solution for the free damped system are not given here for the sake of simplicity, however, the main results are resumed in subsections 3.1 and 3.2. The interested reader is referred to the book by Diana and Cheli for a mathematical justification of the statements reported below until the end of this section.

3.1 Low damped systems

The form taken by the general solution of equation (3.1) is different depending on the amount of damping introduced in the system by the damping matrix $[c]$. Therefore, in this sub-section and the next one, a distinction is made between “low damped” systems and systems containing a “larger”

amount of damping. It is not yet possible at this time to state a quantitative criterion to state, given the parameters of a multi-dof system, if that system should be considered as “low damped” or not, so for the time being this distinction is left as qualitative. After having introduced the principle of modal superposition, it will be possible to provide a quantitative criterion to discriminate between low damped and heavily damped systems.

If the system contains a “low” amount of damping, the $2n$ exponents λ will take the form of complex conjugate pairs, all having a negative real part³:

$$\lambda_{1,n+1} = -\alpha_1 \pm i\omega_1 \quad ; \quad \lambda_{2,n+2} = -\alpha_2 \pm i\omega_2 \quad ; \quad \lambda_{n,2n} = -\alpha_n \pm i\omega_n$$

the corresponding eigenvectors $\underline{x}_0^{(j)}$ will be complex valued.

It is possible to show (see Diana & Cheli for details), that the corresponding general solution for the free damped motion of the system becomes a linear combination of n damped oscillations. Due to the fact that the eigenmodes $\underline{x}_0^{(j)}$ are in this case complex, a different phase is introduced on each single coordinate of the system, so that a representation in matrix form of the general solution is made more difficult. The scalar representation of the solution is the following:

$$\begin{cases} x_{1g}(t) = C_1 |x_{01}^{(1)}| e^{-\alpha_1 t} \cos(\omega_1 t + \angle x_{01}^{(1)} + \varphi_1) + \dots C_n |x_{01}^{(n)}| e^{-\alpha_n t} \cos(\omega_n t + \angle x_{01}^{(n)} + \varphi_n) \\ x_{2g}(t) = C_1 |x_{02}^{(1)}| e^{-\alpha_1 t} \cos(\omega_1 t + \angle x_{02}^{(1)} + \varphi_1) + \dots C_n |x_{02}^{(n)}| e^{-\alpha_n t} \cos(\omega_n t + \angle x_{02}^{(n)} + \varphi_n) \\ \dots \\ x_{ng}(t) = C_1 |x_{0n}^{(1)}| e^{-\alpha_1 t} \cos(\omega_1 t + \angle x_{0n}^{(1)} + \varphi_1) + \dots C_n |x_{0n}^{(n)}| e^{-\alpha_n t} \cos(\omega_n t + \angle x_{0n}^{(n)} + \varphi_n) \end{cases} \quad (3.5)$$

where $|x_{0i}^{(j)}|$ denotes the absolute value (modulus) of the i -th element in the j -th eigenvector, while $\angle x_{0i}^{(j)}$ denotes the phase of the same complex quantity. As can be seen by equation (3.5), there are two main differences in this solution with respect to the general solution of the free undamped motion:

- i) the motion of the system is still the combination of n harmonic oscillations, each one characterised by a specified natural frequency, but in presence of damping these oscillations have a decaying amplitude, that tends to zero for large values of time
- ii) the complex eigenvectors lead to a motion where, considering each one of the n harmonic decaying contributions, each i -th coordinate has its own phase $\angle x_{0i}^{(j)}$. This means that while for an undamped system the motion of two different coordinates will be either perfectly in phase or perfectly in counterphase (depending upon the relative sign of the corresponding elements in the eigenvector), in a damped system each coordinate will reach its maximum value at a different time.

3.2 Systems with high amount of damping

If a multi-d.o.f. system is provided with a high amount of damping, then it is possible that some of the exponents λ obtained as the solutions of equation (3.4) become negative real values.

In this case, we denote by n^* the number of complex conjugate solutions (where obviously $n^* < n$) and we get as solutions of equation (3.4) n^* couples of complex conjugate pairs (with negative real part) plus $2n - 2n^*$ real negative solutions denoted by $-\beta_1, \dots, -\beta_{2n-2n^*}$:

³ As the consequence of $[C]$ being positive definite

$$\lambda_{1,n^*+1} = -\alpha_1 \pm i\omega_1 \quad \dots \quad \lambda_{n^*,2n^*} = -\alpha_{n^*} \pm i\omega_{n^*} \quad \lambda_{2n^*+1} = -\beta_1 \quad \dots \quad \lambda_{2n} = -\beta_{2n-2n^*}$$

thus, the linear combination of exponential solutions in the form (3.2) with the above listed exponents leads quite obviously to n^* exponentially decaying oscillations, with natural frequencies $\omega_1, \omega_2, \dots, \omega_{n^*}$, plus $2n-2n^*$ non oscillatory exponentially decaying motions.

4 Solution for the forced damped motion of the system

As the final part of this lesson, we come to considering the motion of a damped system under the action of time-dependent forces. This problem is governed by a system of equations of motion that takes the “complete” form:

$$[m]\ddot{x} + [c]\dot{x} + [k]x = f(t) \quad (4.1)$$

Forcing effects on structures and mechanical systems may be classified under the following three categories:

- 1) constant forces like e.g. permanent loads on a structure
- 2) time dependent, periodically forces, characterised by the property that:

$$f(t) = f(t+T)$$
 where T is an appropriate time span that is called the “period” of the force.
- 3) transient forces, characterised by the fact that the forcing effect exists only for a finite time duration, and is otherwise zero (this is the case, for instance, of impulsive forces connected with shocks and impacts)
- 4) random forces that are forces that cannot be defined by means of a “deterministic” description, but instead require to be described in statistical terms: typical examples are the excitation applied on a ground vehicle by the roughness of the road, or the force applied by the turbulent wind on an airplane or on a flexible structure.

With respect to the above classification of forces, we observe that:

- 1) Constant forces are by nature conservative, and may thus be included in the right hand side of the equations of motion (in some cases they will indeed disappear) if a proper choice is made on the reference for the independent coordinates, as demonstrated in lesson 1;
- 2) Based on the theory of Fourier series, periodic forces may always be seen as the superposition of several (ideally infinite) harmonic components. Since the equations to be solved (4.1) are linear, using the superposition principle it is possible to obtain the response to any periodic excitation as the superposition of the response to all its harmonic components, each one considered as a single excitation source, which means finding the response to one single harmonic excitation. Thus, the present section will be concentrated on finding the response of the system to a harmonic excitation;
- 3) The response of a linear system to transient forces is instead outside the scopes of this course, since on one side it requires a more sophisticated mathematical treatment and on the other hand the case of transient excitation is less frequent in the engineering practice than the case of periodic excitation. However, this case may be solved by using instruments like the Fourier transform, the Laplace transform and the convolution integral. The reader is addressed to the book by Meirovitch “Fundamentals of vibrations” for details about these techniques.
- 4) Finally the response of a mechanical system to random excitation is a highly interesting topic, because it applies to many relevant engineering applications. However, the theory of random excitation is not simple and requires the use of appropriate mathematical tools (actually, the same already cited in point 3 above) and therefore according to the number of credits allocated to this course will need to be left outside the topics covered by the course.

Owing to the above discussion, it should be clear that by the combined use of the principle of superposition and Fourier series, a very **wide class of forcing effects** (which leaves outside only transient and random forces) may be treated by **computing the response of the system to harmonic excitation**. Therefore, we will confine ourselves to the case of harmonic excitation, that is:

$$\underline{f}(t) = \underline{F} \cos(\Omega t) \quad (4.2)$$

where Ω is the **angular frequency associated to the forcing effect**.

According to the **theory of linear differential equations**, the **general solution to this problem is given by the sum of two functions of time**: the **general solution $\underline{x}_g(t)$** of the corresponding homogeneous set of differential equations **plus the so-called “particular” integral $\underline{x}_p(t)$** :

$$\underline{x}(t) = \underline{x}_g(t) + \underline{x}_p(t)$$

we observe that the component $\underline{x}_g(t)$ of the solution is actually the **general solution of equation (3.1)**, which was **already found in section 3**. Moreover, we observe that, **regardless the amount of damping present in the system, this part of the solution will for sure tend to zero for increasing time** (either through **damped oscillations, damped non-oscillatory motions or both**), and thus will become **negligible after some time**. Thus, $\underline{x}_p(t)$ represents the “stationary” component of the solution.

In order to derive $\underline{x}_p(t)$ we generalise to the case of a multi-d.o.f. system the treatment introduced in Section 1.2. Hence, **we introduce a complex-valued linear differential problem in the form**:

$$[M] \ddot{\underline{x}}^* + [C] \dot{\underline{x}}^* + [K] \underline{x}^* = \underline{F} e^{i\Omega t} \quad (4.3)$$

then, we introduce a **complex-valued solution for this equation**, in the form:

$$\underline{x}^* = \underline{X} e^{i\Omega t} \quad (4.4)$$

deriving with respect to time (4.4), replacing in (4.3) and re-arranging properly we get:

$$(-\Omega^2 [m] + i\Omega [c] + [k]) \underline{X} = \underline{F} \quad (4.5)$$

Note that term $e^{i\Omega t}$ has been simplified from both terms of the equation.

Equation (4.5) represents a **linear, complex-valued, non-homogeneous problem in matrix form that, based on Cramer’s theorem, admits one and only one solution in the form**:

$$\underline{X} = (-\Omega^2 [m] + i\Omega [c] + [k])^{-1} \underline{F} = [G(i\Omega)] \underline{F} \quad (4.6)$$

under the **assumption that the matrix $(-\Omega^2 [m] + i\Omega [c] + [k])$, called the *mechanical impedance matrix*, is non-singular**. The inverse of the mechanical impedance matrix is called the **frequency response function matrix**, and generalises the single FRF defined for the 1-d.o.f. system in Section 2.2. The meaning and structure of the FRF matrix is discussed in Section 4.1 below.

By the way, we observe that **in the case of a positive definite damping matrix $[c]$** , the impedance matrix will for sure be non-singular, indeed its determinant will at least have a non-zero imaginary part. **If instead damping is not present ($[c]=0$) the impedance matrix is singular each time the forcing frequency Ω equals one of the natural frequencies of the undamped system**, as defined by eq. (2.5b). **In this latter case** (that is, a non damped system where the forcing frequency is exactly equal to one of the system natural frequencies), **it may be demonstrated (see Diana & Cheli) that the response of the system is again a harmonic oscillation with the same frequency as the excitation**, but this time the **amplitude of oscillation is linearly increasing with time and thus (at least ideally) tends to infinite for increasing time**.

To complete the solution, we finally take vector $\underline{X}$ from equation (4.7), we replace it into (4.4) and we take the real part of that complex expression. To do so, we observe that $\underline{x}_0$ is a complex-valued

vector, on account of the FRF matrix being complex-valued. Thus, we express the elements in vector $\underline{x}_0$ as complex numbers in polar notation:

$$\underline{X} = \begin{Bmatrix} |X_1| e^{i\psi_1} \\ |X_2| e^{i\psi_2} \\ \dots \\ |X_n| e^{i\psi_n} \end{Bmatrix}$$

Replacing in (4.4) and taking the real part provides the expression for $\underline{x}_p$:

$$\underline{x}_p = \text{Re}(\underline{X} e^{i\Omega t}) = \text{Re} \left\{ \begin{Bmatrix} |X_1| e^{i\psi_1} e^{i\Omega t} \\ |X_2| e^{i\psi_2} e^{i\Omega t} \\ \dots \\ |X_n| e^{i\psi_n} e^{i\Omega t} \end{Bmatrix} \right\} = \begin{Bmatrix} |X_1| \cos(\Omega t + \psi_1) \\ |X_2| \cos(\Omega t + \psi_2) \\ \dots \\ |X_n| \cos(\Omega t + \psi_n) \end{Bmatrix}$$

where we see, as anticipated, that the vibration of the i -th coordinate of the system is described by a harmonic function of time where the angular frequency is the same as for the forcing, the amplitude is defined by the absolute value of the i -th element in vector $\underline{x}_0$ solution of equation (4.6) and the phase of the oscillation is defined by the phase of the i -th element in vector $\underline{x}_0$. Also we see from eq. (4.6) that vector $\underline{X}$ and thus the amplitude and phase of the vibration for each coordinate depend upon the following parameters:

- inertial, elastic and damping properties of the system (matrices $[m]$, $[c]$, $[k]$)
- frequency of the forcing effect Ω .

Finally, we see from the same equation that the amplitude of motion will be proportional to the amplitude of vector $\underline{F}$ representing the intensity of the forcing effect. This last consideration is actually a consequence of the principle of superposition so that, for instance, doubling the amplitude of the force will lead to a double amplitude of displacement.

4.1 The Frequency Response Function (FRF) matrix

We conclude this section and the whole lesson by providing an important physical interpretation of the inverse of the mechanical impedance matrix, called the *Frequency Response Function* matrix. To this end, we observe that since the problem we are treating is linear, the principle of superposition applies, which means:

- i) the overall response of the system to two harmonic force, one denoted by $\underline{F}_i$ acting along coordinate x_i , the second denoted by $\underline{F}_j$ acting along coordinate x_j will be the sum of the response of the system to force $\underline{F}_i$ alone and the response of the system to force $\underline{F}_j$ alone;
- ii) the response of the system to one single harmonic force acting along coordinate x_j and having amplitude $\underline{F}_j$ will be $\underline{F}_j$ times the response of the system to a unit force acting on the same coordinate x_j .

thus, the response of the system to any combination of harmonic forces acting on different coordinates with arbitrary amplitudes may be found by simply combining some “base” functions representing the response of coordinate i to a unit force applied along coordinate j , for all i and j couples.

If we now consider a single unit force acting on the j -th coordinate, the vector $\underline{F}$ representing the intensity of the forcing effect along all coordinates will have a unit value (with zero phase) in position j and will be zero elsewhere:

$$\underline{F} = \begin{Bmatrix} 0 \\ \dots \\ 1 \\ \dots \\ 0 \end{Bmatrix}$$

and correspondingly vector $\underline{x}_0$ representing the amplitude and phase of the stationary vibration of the system will be:

$$\underline{X} = [G(i\Omega)] \begin{Bmatrix} 0 \\ \dots \\ 1 \\ \dots \\ 0 \end{Bmatrix}$$

which gives as a result the i -th column of the inverse of the mechanical impedance matrix.

After this observation, we may state that the i -th element in the j -th column of the FRF matrix $[G(i\Omega)]$ represents (in amplitude and phase) the response of the i -th system coordinate to a unit forcing effect applied along the j -th coordinate. Owing to this physical interpretation, the FRF matrix of the system can be used to define quantitatively the response of the system to any combination of harmonic forces acting on the system.

5 Response to periodic excitation

We consider in this section the response of a mechanical system to periodic excitation. For the sake of simplicity, we address the case of a 1 d.o.f. system, but the theory presented can be easily extended to the case of a multi-d.o.f. system using the concepts introduced in Section 4.

Let us then consider the linear (or linearised) equation of motion for a single d.o.f. system in the form:

$$m\ddot{x} + c\dot{x} + kx = f(t) \quad (5.1)$$

And we assume the time-dependent force $f(t)$ to be any periodic function of time, i.e.:

$$f(t+T) = f(t) \quad \forall t \quad (5.2)$$

with T a specified time interval called the *period* of the force.

According to the theory of Fourier series, any periodic force can be decomposed into harmonic terms, called *harmonics*, as follows:

$$f(t) = F_0 + \sum_{k=1}^{\infty} (A_k \cos(k\Omega_0 t) + B_k \sin(k\Omega_0 t)) \quad (5.3)$$

where:

$$\Omega_0 = \frac{2\pi}{T} \quad (5.4)$$

is called the *fundamental circular frequency* of the periodic force. Hence, according to (5.3) any periodic force can be seen as a composition of an infinite number of harmonic terms, acting at all frequencies which are integer multiples of the fundamental frequency.

Coefficients A_k and B_k in (5.3) can be computed according to the following expressions:

$$A_k = \frac{2}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} f(t) \cos(k\Omega_0 t) dt \quad ; \quad B_k = \frac{2}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} f(t) \sin(k\Omega_0 t) dt$$

whereas F_0 is the mean value of the force:

$$F_0 = \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} f(t) dt$$

Equation (5.3) can also be rewritten in a different form, which is more useful to our purposes:

$$f(t) = F_0 + \sum_{k=1}^{\infty} F_k \cos(k\Omega_0 t + \phi_k) \quad (5.5)$$

with:

$$F_k = \sqrt{A_k^2 + B_k^2} \quad ; \quad \phi_k = \tan^{-1} \left(-\frac{B_k}{A_k} \right)$$

Next, we recall from Section 1.2 that the stationary response of the 1-d.o.f. system represented by Equation (5.1) to a single harmonic excitation having circular frequency Ω is a harmonic motion having the same frequency as the excitation. We also recall that the amplitude of this motion is the modulus of the system's frequency response function (FRF) evaluated at the forcing frequency times the amplitude of the force applied on the system, and that there is a relative phase between the force applied and the resulting vibration, which is the phase of the FRF evaluated at the forcing frequency.

Hence, based on the principle of superposition, we obtain:

$$x(t) = X_0 + \sum_{k=1}^{\infty} X_k \cos(k\Omega_0 t + \psi_k) \quad (5.6)$$

with x_0 the constant displacement of the system under the constant force F_0 :

$$X_0 = G(0)F_0 = \frac{1}{k}F_0 \quad (5.7a)$$

and with:

$$X_k = |G(i\Omega_k)|F_k \quad ; \quad \psi_k = \angle G(i\Omega_k) + \phi_k \quad ; \quad \Omega_k = k\Omega_0 \quad (5.7b)$$

Again using the theory of Fourier series, it is observed that based on equation (5.6) the response of the linear 1-d.o.f. system to a periodic excitation, will also be periodic with the same period T . However, because the modulus and phase of the FRF are not constant with frequency, the waveform of the system's stationary vibration will be different from that of the periodic force applied. This circumstance is sometimes referred to by saying that the system generated a *distortion of the output* (vibration) with respect to the input (force).

Although equations (5.5) and (5.6) are very useful from a theoretical point of view, they would require that infinite harmonic terms are considered to derive the response of the system to periodic excitation. We prefer to look for an approximated solution, by truncating (5.6) to a finite number N of terms:

$$x(t) \cong x_0 + \sum_{k=1}^N x_k \cos(k\Omega_0 t + \psi_k) \quad (5.8)$$

Of course, (5.8) will represent a good approximation of the exact solution (5.6) provided that the overall effect on the solution $x(t)$ of all terms in (5.6) for $k > N$ is sufficiently small. To define a

criterion, we note that the modulus of the FRF is tending to 0 for $\Omega \rightarrow \infty$. Therefore, the FRF modulus is much lower in the seismographic frequency range (frequency of excitation much larger than the system's natural frequency) than in the quasi-static and resonance frequency ranges (frequency of excitation lower than or comparable to the system's natural frequency), as described in Section 1.2.

Therefore, in order for (5.8) to be a good approximation of (5.6) we simply prescribe:

$$\Omega_N = N\Omega_0 \gg \omega = \sqrt{\frac{k}{m}} \quad (5.9)$$

with ω the (undamped) natural frequency of the system.

In conclusion, the response of a linear system (or the small oscillation of a non-linear system around an equilibrium position as described by the linearised equation of motion) excited by a periodic force with period T will be a periodic motion with the same period, and can be seen as the superimposition of a number N of harmonics, whose amplitude and phase are obtained from the amplitudes and phase of the force harmonics based on Equations (5.7a,b).

The process to compute the response of the system to periodic excitation is schematically represented in Figure 2. As it is shown in the figure, this process entails first a transformation of the input (force) from the time domain to the frequency domain (using the Fourier series): this will produce what is called the force spectrum. Then, using the system's FRF the spectrum of the periodic response of the system can be obtained in the time domain. Finally, the solution is transformed to the time domain, using again the Fourier series.

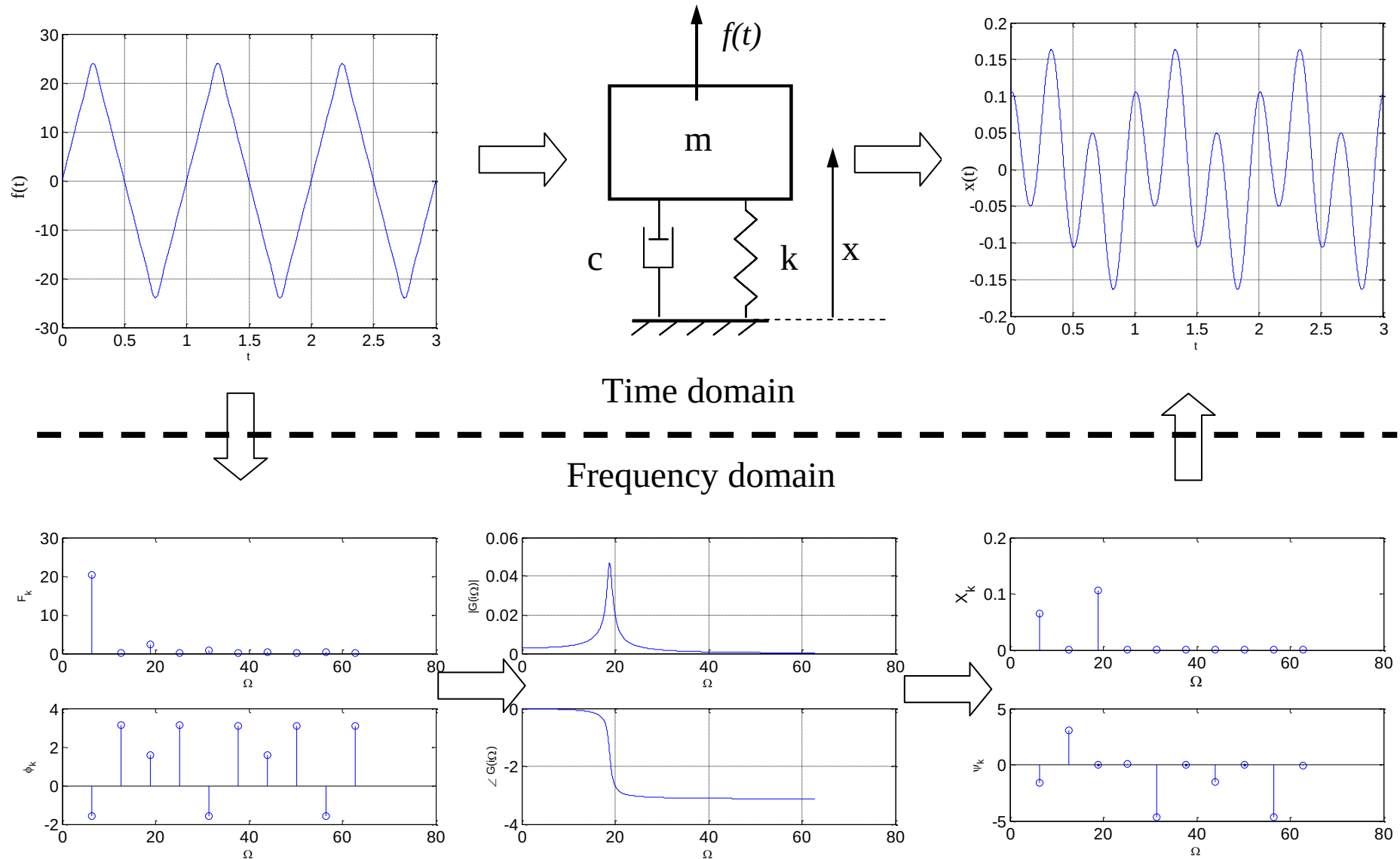


Figure 2: graphic representation of the calculation of the response of a vibrating system to periodic excitation